AI & ML Internship – Task 2
Feature Engineering, Model Optimization & Performance Comparison

This report presents the implementation of an enhanced House Price Prediction system using Machine Learning techniques. The project demonstrates preprocessing, feature scaling, model training, performance evaluation, and model comparison using the California Housing Dataset.

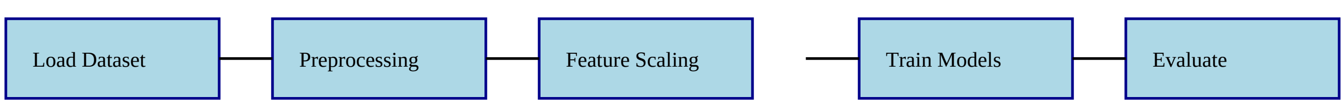
Project Objectives

Objective	Description
Dataset Analysis	Explore and understand housing-related data
Preprocessing	Apply feature scaling and train-test split
Model Training	Train multiple regression algorithms
Performance Comparison	Compare metrics to identify best model
Model Selection	Choose the highest-performing model

Dataset Used

The California Housing Dataset contains real-world housing-related attributes including: Median Income House Age Average Rooms Population Latitude and Longitude Target Variable: **Median House Value**

Machine Learning Workflow



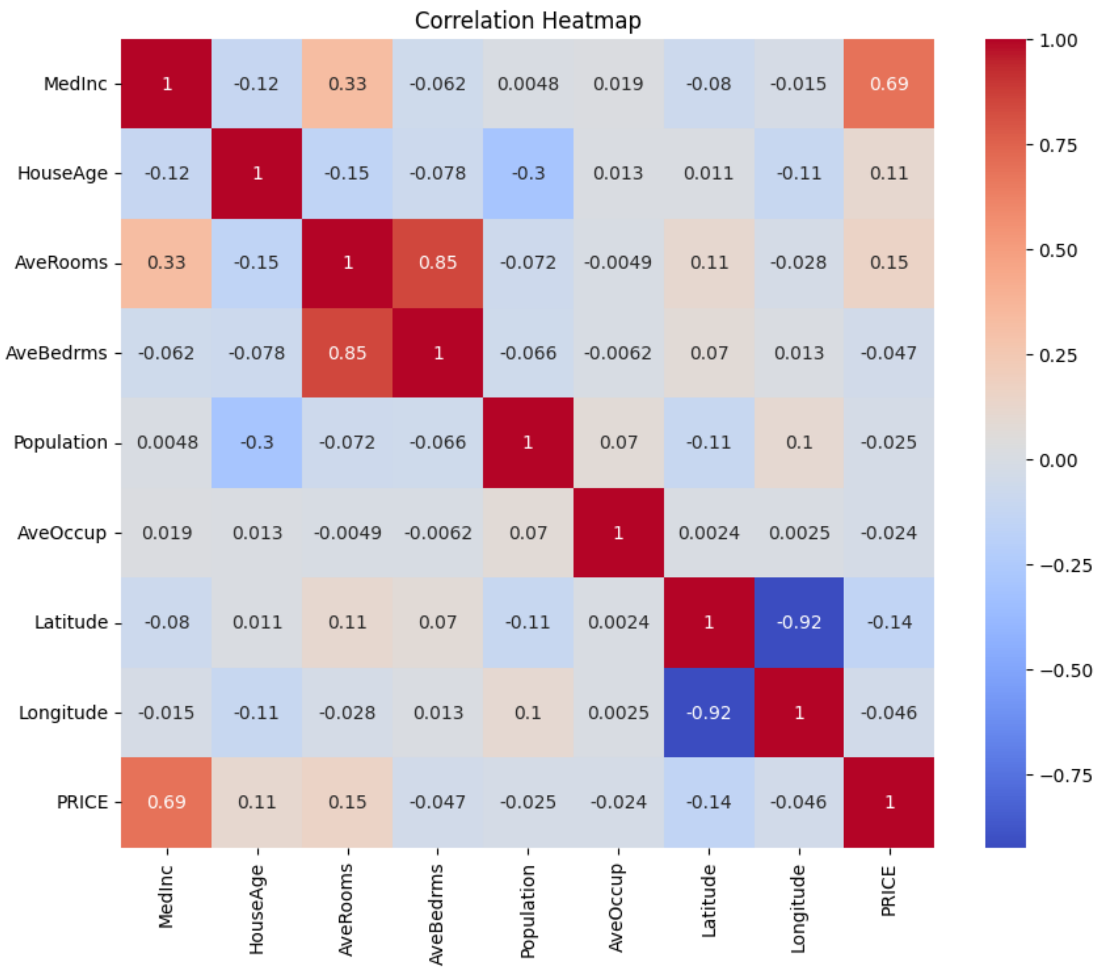
Methodology

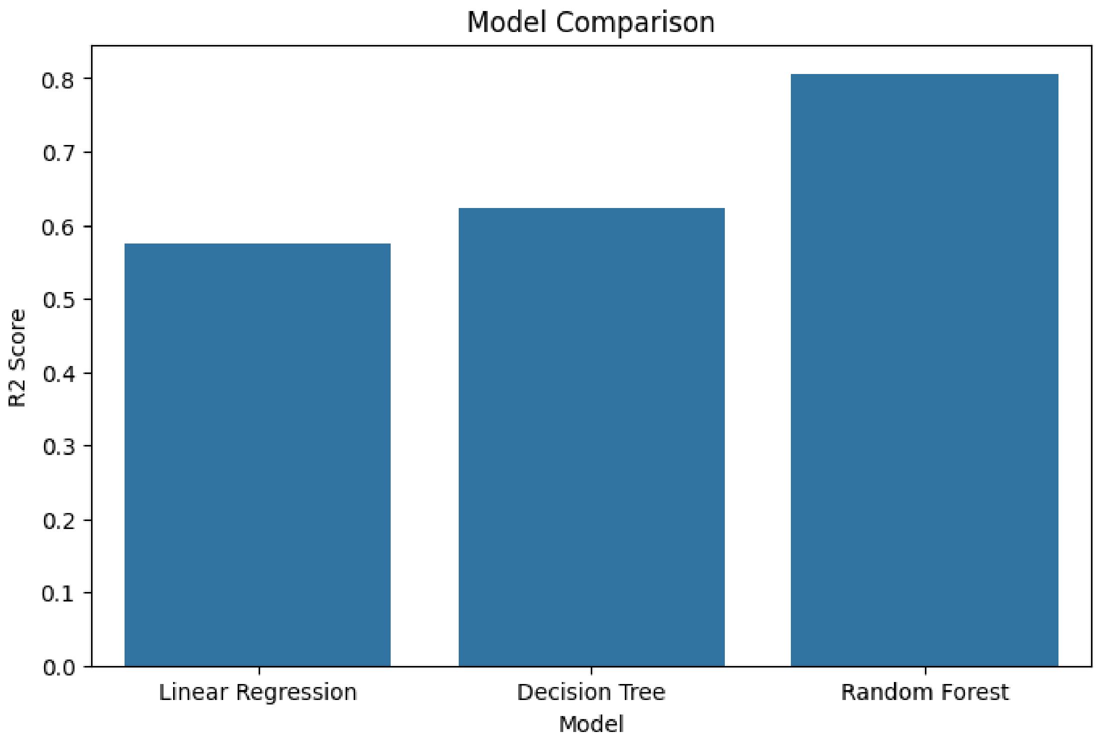
The following Machine Learning workflow was implemented: Imported and explored the California Housing Dataset Performed data preprocessing and visualization Applied StandardScaler for feature scaling Split data into training and testing sets Trained Linear Regression, Decision Tree, and Random Forest models Evaluated models using MAE, MSE, RMSE, and R^2 Score

Evaluation Metrics

Metric	Purpose
MAE	Measures average prediction error
MSE	Measures squared prediction error
RMSE	Measures root mean squared error
R^2 Score	Measures overall model accuracy

Notebook Visual Outputs





Model Performance Summary

Model	Expected Performance
Linear Regression	Moderate Accuracy
Decision Tree Regressor	Good Performance
Random Forest Regressor	Best Overall Performance

Conclusion

This project successfully demonstrated a professional Machine Learning workflow including: Data preprocessing Feature scaling Regression model training Performance evaluation Model comparison and optimization Among all models tested, the Random Forest Regressor achieved the best performance and was selected as the final model for House Price Prediction. The project provided practical understanding of industry-aligned Machine Learning practices.